



**THE INFLUENCE OF THE SEROTONIN TRANSPORTER ON
NEUROGENESIS.**

**A STUDY INTO THE PATHOGENESIS OF DEPRESSION AND ANXIETY-
RELATED DISORDERS**

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Abstract

INTRODUCTION: Decreasing the serotonin transporter (SERT) functioning is the primary target of therapies in depression and anxiety-disorders. In adulthood this results in a reduction of symptoms. But decreasing the functioning in early life enhances the susceptibility to psychopathology in later life. A possible explanation for this phenomenon is neurogenesis, as neurogenesis is influenced by serotonin concentrations and almost extinct in adulthood. This study examines the influence of the SERT genotype and the maternal SERT genotype on neurogenesis in rats.

METHODS: Rats with a genetic mutation causing a reduced SERT expression were used for this experiment. Heterozygous (HET) males were bred with homozygous (HOM), heterozygous and wild type (WT) females to create seven different possible combinations of genotypes. BrdU immunohistochemistry was used to quantify the neurogenesis. Rats were injected with BrdU on post-natal day (PN) 7 and sacrificed 2 hours later. Brains were stained to reveal the BrdU epitopes, which were counted in the amygdala and hippocampus. Apart from the BrdU staining, also the bodyweight was measured on PN 7.

RESULTS: There were no results regarding the BrdU analysis. A personal mistake by the writer of the thesis made it impossible to see the BrdU staining. Analysis of the bodyweight showed a significant difference for maternal genotype ($p = 0.002$) but not offspring genotype ($p = 0.253$). The difference was caused by a significant decrease in bodyweight in HOM offspring compared to WT offspring ($p = 0.002$) and a non-significant decrease, but a strong tendency, in HOM offspring compared to WT offspring ($p = 0.056$).

DISCUSSION: Results of the neurogenesis study remain to be investigated and cannot be discussed at this moment. Prenatal- and postnatal effects could cause the found difference of the pup bodyweight. Future studies are needed to investigate the cause of the difference in bodyweight.

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Introduction

Depression and anxiety-related disorders are prevalent diseases in contemporary society and have potentially long-lasting effects. On the global top 10 of years lived with disability, depression occupies the third place, with anxiety disorders at nine (1). With approximately 615 million people suffering from depression or anxiety, these conditions account for 30 percent of the total global non-fatal disease burden (2). Lifetime prevalence rates are 9.6 percent for developing major depressive disorder and 12.9% for developing an anxiety related disorder (3). Leading to an annual cost of 74.4 billion dollars for anxiety disorders and 113.4 billion for mood disorders in Europe (4).

The pathophysiology of psychiatric disorders is yet to be fully elucidated. However, mounting evidence suggests that depression and anxiety have early developmental origins (5). The lack of knowledge makes the need to understand the changes that occur during the brain's development in order to unravel the pathophysiology and pathogenesis of psychiatric disorders more pressing.

Serotonin has long been implicated as a crucial molecule in the development of psychiatric disorders. Serotonin is most well known as a neurotransmitter involved in the regulation of mood, appetite, social and sexual behaviour, cognition, attention and arousal (6–8). Furthermore, serotonin's role as a trophic factor has long been established. It is known to possess the ability to modulate developmental processes such as neurogenesis, apoptosis, axon branching and dendritogenesis (8,9). Maintaining serotonergic homeostasis is a delicate process, but it is crucial in the functioning of the brain. Disruptions of serotonin homeostasis is linked to the development of several psychiatric disorders, including schizophrenia, affective disorders, depression, anxiety and autism (7,10–12).

A key factor in the serotonin homeostasis is the serotonin transporter (SERT). The SERT is a transmembrane protein that regulates the extracellular concentrations of serotonin (13). It does so by taking up serotonin into the presynaptic neurons, where it can be reused or converted into inactive molecules. An increase in the expression of SERT leads to a decrease in the extracellular

concentration of serotonin and vice versa. Changes in SERT expression have the most significant influence on extracellular serotonin concentrations (14). SERT expression can be disturbed by genetic, environmental and pharmacological factors.

The SERT is also a main target in therapies intended to address depression and anxiety-related disorders. The most common approach to therapy involves drugs called selective serotonin reuptake inhibitors (SSRIs), which are the most frequent prescribed and first-choice treatment for depression and generalized anxiety disorder (15). They work by selectively blocking the serotonin transporter on the presynaptic neuron, thereby decreasing intracellular concentrations of serotonin and increasing extracellular concentrations (7). These drugs have demonstrated clinical efficacy in relieving the symptoms of depression and anxiety-related disorders when administered in adulthood (16).

The role of genetics in depression and anxiety disorders

Depression and anxiety-related disorders are both suspected to have a genetic component. Homozygous twin studies have shown an inheritability of depression and anxiety disorders of approximately 30 to 40 percent (17,18). In addition, genetic epidemiological studies suggest a genetic component in the pathophysiology of depression (18,19). While the study of genetics have proven to be a valuable tool in predicting the pathologies and progressions of numerous diseases, the relationship between depression and genetics remains elusive. The pathophysiology of depression involves a complex interplay between genetic and environmental factors; nonetheless, a number of genes have been identified that could account for, or at least participate in the pathophysiology of depression.

The majority of existing research into the influence of genetics on the pathophysiology of depression has focussed on the *SLC6A4* gene. The *SLC6A4* gene is situated on chromosome 17 and is the only gene in the human body that codes for the production of SERT(20,21). Multiple mutations in the *SLC6A4* gene have been discovered over time. While the majority of these are very rare single-nucleotide mutations (SNPs), that have no relation to SERT functioning, there are, however a number of polymorphisms that have a significant effect on SERT activity. In the promotor region, there are

two polymorphisms of interest, namely the so-called 5-HTTLPR, the rs25531. In intron 2 there is another relevant mutation called the STin2 polymorphism. The 5-HTTLPR is a variable-number of tandem repeat mutation that leads to the insertion or deletion of 44 base pairs in the promotor region of the SLC6A4 gene. The genetic variant in which the 44 base pairs are deleted is known as the short allele (s-allele), while the genetic variant without the deletion is known as the long allele (l-allele).

Beyond genetic differences, a change in promotor length also causes changes in SERT expression. In humans, individuals who express two short alleles (ss-carriers) or one short and one long allele (sl-carriers) show reduced SERT expression, when compared to those with two long alleles (ll-carriers)(22). A reduction in SERT presence presumably leads to an increase in extracellular serotonin concentrations. The 5-HTTLPR polymorphism is common in the Caucasian population, as approximately 70 percent of the Caucasians have one or two copies of the s-allele(23). This makes the impact of the behavioural effects caused by the 5-HTTLPR polymorphism on society immense.

While the 5-HTTLPR polymorphism is the most extensively researched mutation, there are others that influence SERT expression. One of the other polymorphisms of interest is the rs25531 polymorphism. This polymorphism is a single nucleotide mutation in which a guanine residue replaces an adenosine residue. The rs25531 polymorphism is almost always in phase with the l-allele; it causes a reduction SERT production in a manner similar to that of the ss-carriers. Therefore, an additional distinction is made between ll-carriers with the rs25531 polymorphism (L_A -carriers) and without (L_G -carriers). In articles analysing the effect of the 5-HTTLPR polymorphism, this additional distinction between the L_A - and L_G -carriers is frequently ignored, resulting in underestimations of the real-world impact of mutations in the 5-HTTLPR gene(24). Especially if different ethnical groups are analysed the extra distinction is crucial. The L_A genotype is more common in Caucasians compared to African Americans(25). The final mutation of interest is STin2 polymorphism. STin2 polymorphism causes a variable-number tandem repeat mutation that affects SERT transcription. The exact biological consequences of this polymorphisms are unknown, but STin2 polymorphism with a larger number of repeats has been associated with an increase in SERT transcription(26).

The 5-HTTLPR polymorphism has also been shown to be involved in several behavioural phenotypes. The literature is contradictory, but the majority of published studies and meta-analyses indicate a relationship between the 5-HTTLPR polymorphism, early life adverse events and the development of depression later in life (27,28). Furthermore, the ss- and sl-carriers have been shown to be associated with more anxiety related behaviours, such as neuroticism and negative emotionality(29). This indicates that it is likely that genetic mutation of the SERT play a role in the pathophysiology of mental disorders.

Furthermore, the 5-HTTLPR polymorphism has been shown to impact the morphology of the human brain. The possession of an s-allele results in enhanced hypothalamic–pituitary–adrenal axis (HPA-axis) activity, which could account for an extensive stress response to stressful stimuli. Over-activity in the HPA axis is suggested to be related to the amygdala, which has been shown to respond excessively to stressors in s-carriers, which alters the resting state of the amygdala (30–32). A reduction of functional connectivity between the anterior cingulate cortex and the amygdala is also noted (30). Due to the inhibitory effect of the cingulate cortex on the amygdala it has been suggested that this reduction could explain the increased stress response of ss-carriers (30,33). The impact of the 5-HTTLPR polymorphism on the hippocampal volume is contradictory: In ss-carriers and sl-carriers with major depressive disorder, both increased and decreased hippocampal volume have been described(34).

Genetic animal models

The 5-HTTLPR polymorphism does not exist in rodents, but it can be mimicked in the genetically modified SERT knock-out (SERT^{-/-}) mice and rats. The most commonly used technique for creating SERT^{-/-} in mice is homologous recombination, which was developed in 1999(35). The SERT knock-out rat was not available up until 2006. It was created by means of targeted induced local lesions in genome (TILLING) using the N-ethyl-N-nitrosourea (ENU) method. The ENU method creates random point mutations in spermatogonial stem cells after intraperitoneal injection(36). The DNA of the offspring is scanned for mutations of interest. Several different genetically altered rats have been created using this technique. One of the rats had a premature stop codon in the SERT gene, causing a

total ablation of the SERT production. The rat with the changed SERT gene was mated with wildtype rats for several generations in order to outcross other possible mutations(36). SERT concentrations were measured in vivo; it was found that SERT had disappeared completely in SERT^{-/-} animals and was reduced to 60 percent compared to the original concentrations in SERT-heterozygous (SERT^{+/-}) animals(37). The deletion of the SERT affected the extracellular serotonin concentrations in the mutant rats. Analysis of the extracellular serotonin levels in the rats' hippocampi showed a nine-fold increase in SERT^{-/-} animals compared to SERT^{+/+} animals; this increase in serotonin levels is similar in both SERT mice and SERT rats(33). On the other hand, intracellular serotonin levels were found to have been reduced by approximately 65 percent in SERT^{-/-} rats. These observations indicate reduced SERT activity and therefore a reduction in neuronal serotonin reuptake.

The serotonergic changes in the SERT^{-/-} rats also resulted in a different behavioural phenotype: SERT^{-/-} animals showed an increase in anxiety and depression-related behaviours (22,38,39). Olivier et al. found an increase in anxiety using the open-field test, the elevated plus maze test and the home-cage emergence test(40). They also found an increase in depressive behaviour in the forced swim test and measured an increase in anhedonia-like behaviour measured with the sucrose consumption test. These results make the SERT^{-/-} animals a well-established model for assessing depression and depressive-like behaviour (40).

Behavioural abnormalities are not the only differences discovered in SERT^{-/-} rodents, as morphological brain development is also altered by decreased SERT expression. Changing serotonin concentrations led to abnormal development of the whisker-barrel fields in the somatosensory cortices in mice (41). In addition, Wellman et al. showed morphological abnormalities in the dendritic branches of the pyramidal neurons of the limbic cortex and discovered an increase in spine density in the basolateral amygdala in SERT^{-/-} mice. The amygdala has been studied extensively in combination with the SERT^{-/-}. Nietzer et al found that SERT^{-/-} mice demonstrate an increase in the total number of spines and the total number of spines per μm in the basolateral amygdala, as well as an increase in the number of branches per 100 μm and a total number of spines per 100 μm in the lateral amygdala(42,43). The amygdala was also found to have reduced changed connectivity to the

prefrontal cortex, resulting in decreased inhibitory feedback provided by the amygdala which likely could contribute to the increased responses to stressful events (43,44). An increase in stem cell proliferation rates in the hippocampus has been measured in aged SERT^{-/-} mice (14.5 months) but not adult mice(45). The SERT mutation has been shown to alter concentrations of brain-derived neurotrophic factor (BDNF). SERT^{-/-} rats show a reduction in BDNF expression, which is a crucial molecule in the formation of cortical structures and suspected to be involved in depression(46). The 5HT1a receptor density is changed in SERT^{-/-} animals as compared to SERT^{+/+} animals. The 5HT1a receptor is desensitized by increased serotonin concentrations; these disruptions are suspected to be key in the modulatory changes that occur in SERT^{-/-} animals (47,48).

Maternal influence

During the perinatal period, serotonin concentrations are not only affected by serotonin metabolism within the individual but also by the maternal environment(49). Serotonin is one of the earliest available neurotransmitters in the brain. Serotonergic neurons are available as early as the fifth week during the human gestation and in mice at embryonic day (E) 10 of the total 28 days(50). However, serotonin, serotonin receptors and SERT are already found in the mouse brain before E10, suggesting a placental source for serotonin(8,51,52). Serotonin cannot pass the placental barrier or the blood-brain barrier, but Bonnin et al. suggested a mechanism for the regulation of the serotonin concentrations in offspring that is based on tryptophan concentrations in the mother. Tryptophan is the precursor of serotonin and concentrations of tryptophan and serotonin are correlated. Thus an increase in tryptophan means an increase in serotonin concentration. Tryptophan can pass the placenta and, in the placenta, is partly converted into serotonin by aromatic L-amino acid decarboxylase. Based on this phenomenon, maternal serotonin concentrations in early embryonic development could be a dependent factor in the development of a child (49,53). Muller et al. provide evidence for this theory: In a mouse model with SERT over expression, they found changed serotonin concentrations in the forebrain at E14.5 depending on maternal genotype. The serotonergic neuron network is not fully active before E14.5 thus, this changed concentration is likely a result of maternal contribution (52,54). The exact effects of the maternal genotype remain undiscovered, but human research also indicates

that maternal genotype has an influence. Van der Knaap et al. found a role for maternal 5HTTLPR genotype in behavioural and brain morphology in humans. They identified a difference in behavioural performance in a visuomotor precision task and the morphology of the somatosensory cortex. The authors provided two possible explanations for their findings. The first is a difference in the environment during prenatal development, while the second is based on postnatal development, meaning that maternal behaviour could differ according to genotype(55).

The SSRI paradox

Serotonin is a molecule that has a simple structure but a wide array of functions. One of the most notable curiosities of serotonin is the difference between its effect in early life and in adulthood. Enhancing serotonin concentration is the primary goal of the majority of anti-depressant therapies, as doing so produces beneficial effects relieves depressive and anxiety-related symptoms (16). However, treatment with SSRIs has been shown to increase extracellular serotonin and decrease the intracellular serotonin levels in a manner similar to that of the SERT mutation(45,56). Furthermore, as described in the previous paragraphs, increasing the serotonin concentration during early development causes the exact opposite effect of SSRI treatment in adulthood. The differing effect of high serotonin concentrations during different life phases is known as the SSRI paradox. The cause of this paradox remains elusive.

Neurogenesis

One of the mechanisms that could be partly responsible for the SSRI paradox is the influence of serotonin on neurogenesis. Neurogenesis is the formation of new neurons from neuronal precursors. The process is crucial for the development of the brain(7). Disruptions to prenatal neurogenesis have been shown to influence the development and size of the brain, resulting in brain-related disorders such as intellectual disability (mental retardation), autism spectrum disorders, childhood-onset schizophrenia, epilepsy, paediatric bipolar disorder, attention-deficit hyperactivity disorder and microcephaly (57,58).

The vast majority of the neurogenesis occurs during the prenatal and early postnatal period. In adulthood, neurogenesis is almost extinct. There are only a small number of areas of the brain that still show neurogenesis throughout all periods of life. Two of the most prominent brain areas that show adult neurogenesis are the dentate gyrus (DG) in the hippocampus and the subventricular zone(59,60). Adult neurogenesis is a complex process that is affected by multiple factors, and it has been suggested that it is involved in the pathophysiology of depression(61). This suggestion is based on the discovery that neurogenesis rates are reduced in depressed patients when compared to healthy individuals. There is no direct measure of neurogenesis in humans but post-mortem human studies and MRI volumetric studies have found smaller dentate gyrus sizes in depressed individuals, which could be the consequence of reduced adult hippocampal neurogenesis(62,63). Animal research provides deeper insight into the role of neurogenesis in depression. Rodent studies show a decrease in the rate of neurogenesis as a response to stress and in behavioural animal models of depression (64,65). This decrease can be reversed by treatment with anti-depressants (60). Another indication that for neurogenesis is involved in depression is the late onset effect of anti-depressant therapy. On average anti-depressant therapy takes four to six weeks before notable improvement takes place. This is similar to the period of time that new neurons require to mature and be integrated into the neuronal network in the DG (60).

Linking this back to the role of serotonin, serotonin is an important factor in neurogenesis (7). Depletion of serotonin has been shown to decrease the adult neurogenesis rates, while increased concentrations of serotonin cause the opposite effect (60,66). In light of the SSRI paradox, serotonin concentration is only elevated in adulthood with SSRI treatment. At this point in time serotonin only affects the areas that still show neurogenesis in adulthood. Elevated serotonin concentrations in early life may however, increase the neurogenesis rate all throughout the brain, thus leading to changes in brain structure and therefore the enhanced susceptibility to psychopathology.

Analytical technique in this study

One of the techniques used to analyse neurogenesis is Bromodeoxyuridine (BrdU) immunohistochemical staining. BrdU is an exogenous thymidine analogue that is administered by

intraperitoneal injection. In vivo, BrdU is incorporated into the DNA during the synthesis phase (S-phase) of the cell cycle(67). The difference between thymidine and BrdU lies in the bromide group depicted in Figure 1. After incorporation, cells expressing BrdU can be detected in the DNA using

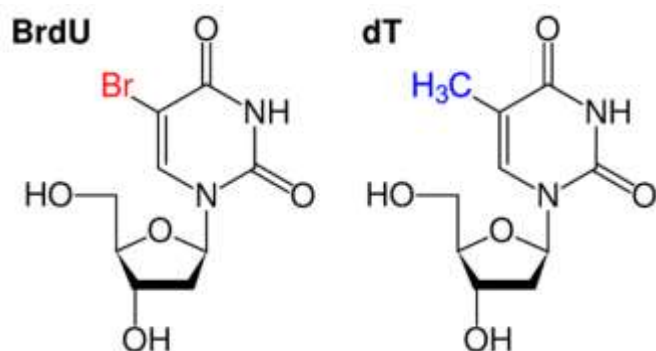


Figure 1: The chemical structure of BrdU and Thymidine

immunohistochemistry techniques. The target of immunohistochemical labelling is single-stranded DNA containing BrdU(67). Depending on the concentration, frequency and duration of the BrdU administration, it can be used for several neuronal experiments.

One of the most common uses of BrdU is marking newly formed cells in living tissue. It is a fast-working molecule and, in the right dosage can mark all proliferating cells. In this thesis, BrdU is used to follow the formation of new cells in brain tissue, focusing specifically on neurogenesis.

Neurogenesis can be examined using BrdU during a short interval between injection and analysis. The half-life of BrdU in plasma is approximately 11 minutes (68). Labelling in the DNA occurs in the first two hours after administration, after which availability drops rapidly. The entire cell cycle takes approximately 12 hours, of which the S-phase spans a third. To measure only the formation of new cells, an animal should be sacrificed two hours after a BrdU injection. If the time interval is longer, the newly formed neurons might migrate to other brain areas (69).

This thesis

This study is designed to present a possible explanation of the SSRI paradox and to gain deeper insight into the underlying pathophysiology of depression. The main objective of the study is to investigate the influence of SERT genotype on neurogenesis in the hippocampus and the amygdala of male rats at postnatal day (PN) 7. As a secondary goal, the relationship between offspring genotype and maternal genotype on the neurogenesis was investigated. Based on the previously described

literature and a pilot study conducted in the lab (data not published), the expectation is that increased extracellular serotonin concentrations will lead to an increase in neurogenesis. Therefore, the homozygous (HOM) rats are expected to express the highest neurogenesis rates. The role of the maternal genotype is expected to be rather more complex. Due to of the experimental setup used, the rats were analysed at PN 7; thus, no direct placental transport of tryptophan from the mother was possible in the interval after BrdU injection. Therefore the influence of the maternal genotype should largely be restricted to prenatal morphological changes as a result of different serotonin concentrations or behavioural changes in mother-pup interactions based on genotype. Because of the complexity of the neurogenesis process it is difficult to predict the effect of the maternal genotype. An increase and decrease of neurogenesis rates are both possible.

Beyond the role played by the SERT in neurogenesis, the roles played by the SERT and maternal genotype in several other variables are also examined in this thesis. These other variables are the surface of different brain areas and the bodyweight of the rats.

Methods

Animals

Wistar rats were used for all the experiments. The SERT mutation was originally created by ENU- driven target-selected mutagenesis, causing a premature stop codon in the *SLC6A4* gene that lead to a total ablation of the production of SERT(36). The rats were held at $21 \pm 2^\circ\text{C}$ and a constant humidity of 55 percent with a day-night cycle of 12 hours (the lights were off between 7am and 7 pm). Ad libitum food and water were available. Only the male rats were used, as time constraints made it impossible to obtain a sample size large enough to account for the differences between males and females. The choice of males was due to of the tendency to use male animals in previously published literature, thus increasing the comparability of the results of this work with previous findings. Heterozygous (HET) males were bred with HOM, HET and wild type (WT) females to create seven different possible combinations of genotypes (Figure 2). To correct for inter-litter bias, multiple litters per maternal genotype were used. All the procedures were approved by the Victoria University of Wellington Animal Ethics Committee (application ID: 0000022375).

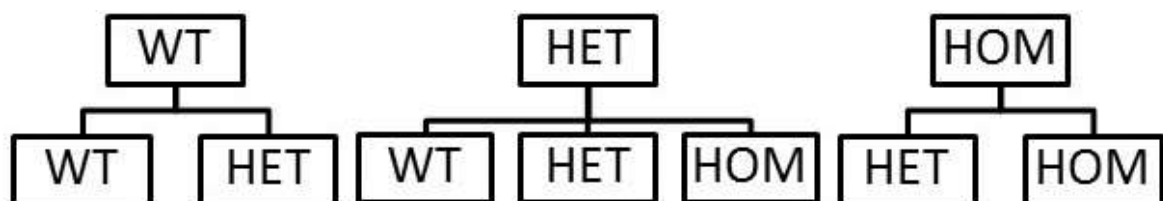


Figure 2: Possible genotype combinations. The first row shows the possible maternal genotype, while the second row shows the possible genotypes of the offspring.

Brain preparation

On PN 7, the pups were taken from their home cages and brought over to another room and weighed, after which they were injected intraperitoneal with 25 mg/kg BrdU, based on previous pilot studies in our laboratory. The pups were taken two at a time, and the total time from separation from the mother to return was no more than five minutes. The pups were marked on their backs for identification purposes. Two hours after injection, the animals were deeply anaesthetized with

50mg/ml pentobarbital: the amount administered was 100 mg/kg i.p.. When an animal demonstrated no visible response to a pinch to the feet, it was quickly decapitated. The brains were removed and tail clips were taken for genotyping. Genotyping was performed by Transnetyx (Cordova, USA) and the tail clips were scanned for SERT genotype. The brains were immersed in a 4 percent paraformaldehyde solution for 24 hours and transferred to a 30 percent sucrose solution thereafter. The brains were kept in the sucrose solution up until the point that they were fully immersed in sucrose, after which they were snap-frozen in isopentane at -40 °C. The snap-frozen brains were stored at -70 °C up until slicing.

Tissue preparation

The tissue was prepared with a protocol derived from a study conducted by Steel et al. (70). Following the freezing, coronal 30 µm brain slices were prepared on a freezing microtome. Every sixth slide containing a section of the hippocampus or the amygdala was used for staining. In order to prevent counting the same cell twice, the difference between different brain parts was 180 µm. Staining was performed using the avidin-biotin method, with a diaminobenzidine reaction. Antigen was retrieved by incubating with 2N HCL for 90 minutes after which the non-specific binding sites were blocked by incubation with 1 percent bovine serum albumin (BSA) for one hour. Samples were incubated overnight with the primary anti-BrdU mouse monoclonal antibody conjugated to biotin (ab2284, Abcam, stored at -80°C). The next day, the samples were incubated with 0.1% ABC peroxidase complex (Vectastain Elite Kit, stored at 4°C) for 60 minutes and then with 3,3-diaminobenzidine (DAB) for 20 minutes in order to reveal the BrdU epitopes. Thereafter, the slides were counterstained with neutral red and mounted under DPX (a mixture of di styrene, tricresyl phosphate and xylene).

Slide analysis

All of the slides were coded and randomised by an independent investigator to ensure objectivity. A total of four slides per brain were analysed. The slides were matched according to Bregma coordinates to obtain consistency in coronal area analysis. The areas of the hippocampus (Cornu

Ammonis (CA) 1/2, CA 3 and the dentate gyrus) and the amygdala were identified according to the atlas of the developing rat brain by Khazipov et al. (71). The surface of the regions was measured (μm) and cells in these regions were counted manually. Cell density was used as the primary variable of interest.

Data analysis

Data analysis was performed using IBM SPSS Statistics 24. The data obtained from weighing the pups, as well as the data provided by the neurogenesis experiment, were analysed. Weight was analysed using the following tests: A one-way ANOVA was used to assess the influence of genotype on weight. A one-way ANOVA was used to assess the relationship between maternal genotype and weight. Finally a two way factorial ANOVA with interaction effects was used to see if there was an interaction effect between offspring and maternal genotype. Post hoc testing was performed using Tukey's test. The fixed factors for the two-way ANOVA were pup genotype and maternal genotype, while the dependent variable was weight. A subgroup analysis was conducted using a one-way ANOVA with only HET offspring to further assess the influence of maternal genotype on weight. The effect of litter size on weight was analysed using linear regression analysis and Pearson's correlation test.

The data from the neurogenesis study were analysed for different variables, using the same ANOVA tests as described for the weight analysis. The primary variable of interest was cell density, in cells/ μm^2 . The data from the six slides of a single mouse brain were combined and analysed together. Other variables tested were the sizes of the different brain areas, corrected for the weight of the animals. Furthermore the effect of weight on the cell density was analysed by linear regression analysis and Pearson's correlation test.

Results

Animals

In total thirteen litters were used, of which six were of the HET maternal genotype, three of the WT maternal genotype and four of the HOM maternal genotype. In total the thirteen litters consisted of 147 animals of which 47 percent were male resulting in 67 male animals being used for the experiment. The genotyping of the offspring showed 17 HOM pups 31 HET pups and 19 WT pups. Further details concerning the genotypes are provided in Table 1.

Table 1. The animals used in this experiment

	Offspring Genotype				
Maternal Genotype		HOM	HET	WT	Total
	HOM	10	9	N/A	19
	HET	7	10	10	27
	WT	N/A	12	9	21
	Total	17	31	19	

Weight analysis

On PN day 7 the pups used for the neurogenesis experiment were weighed before injection with BrdU (see fig 3). According to the Shapiro-Wilks test, the data obtained from weighing the pups showed a normal distribution. Furthermore, Levene's test showed no difference in variance in any of the tests.

The data were analysed using the different ANOVAs. This was necessary in order to correct for the imbalance in the experimental design, in particular the unequal distribution of offspring genotypes per maternal genotype (see figure 2). The first analysis with a two-way ANOVA showed an average weight over all groups of 15.1 ± 1.87 g. There was no significant difference based on offspring genotype ($F(2,64) = 1.405$, $p = 0.253$ and $\eta^2 = 0.042$); however, there was a significant effect for maternal genotype ($F(2,64) = 6.822$, $p = 0.002$ and $\eta^2 = 0.176$). There was no interaction effect between maternal genotype and offspring genotype ($p = 0.481$). Post hoc analysis with Tukey's

post hoc test showed a significant decrease in bodyweight in HOM offspring compared to WT offspring ($p = 0.002$) and a non-significant decrease, but a strong tendency, in HOM offspring compared to WT offspring ($p = 0.056$).

Although the interaction itself was not significant, exploratory analysis of the body weight of the HET offspring using a one-way ANOVA, showed a significant difference ($F(2, 28) = 5.31$ $p = 0.011$ and $\eta^2 = 0.275$). A Post Hoc Tukey's test showed a significant increase in HET offspring from WT mothers compared to HET offspring from HOM mothers ($p = 0.011$), HET offspring from HET mothers compared to HET offspring from HOM mothers ($p = 0.047$) but no difference between HET offspring from WT mothers and HET offspring from HET mothers ($p = 0.86$). Indicating a maternal effect. No difference was found for maternal genotype when compared to litter size or correlation between litter size and weight.

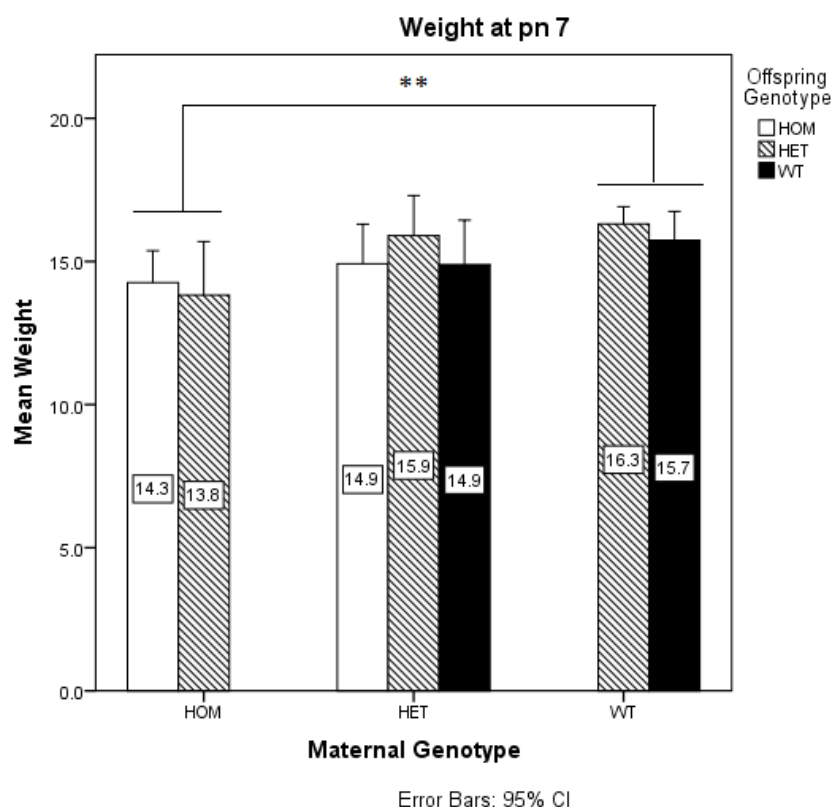


Figure 3: The influence of maternal SERT genotype and offspring SERT genotype on the weight of the rat pups at PN 7.

Error bars represent the 95% CI. ** = $p < 0.01$

Results of the neurogenesis analysis

Because of a personal mistake made by the writer of this thesis, the results were not available at the point in time when the internship ended. Due to the fact that the internship took place in New Zealand, an extension of the research period was not possible. The experiments were to be completed over the following months by a fellow lab member, but due to an unknown reason this did not take place. Thus, the results of the neurogenesis experiment remain to be investigated.

The mistake described above should not have occurred in the first place. However, due to an interpretation mistake while executing the staining of the brain slices the concentration of hydrogen peroxide was too high during the DAB staining. The DAB was intended to bind the peroxidase complex attached to the BrdU antibodies. Extra hydrogen peroxidase was added to react with the excess of DAB. Because of the high concentration of hydrogen peroxide, it seems reasonable to suspect that all the DAB reacted with the hydrogen peroxide. Therefore, no staining was visible when examining at the slides under a microscope.

One batch of brain slices was treated with the correct concentration of hydrogen peroxide. In this batch, the DAB-stained cells were visible on all of the slides. Thus, it seems clear to state that the problem in the analysis resulted from the concentration of hydrogen peroxide.

Discussion

Weight analysis

The analysis of the male pup weight at PN 7 showed an effect for maternal genotype but not for offspring genotype. Pups from HOM mothers caused the effect, as they were significantly lighter than pups from HET or WT mothers. Furthermore, analysis of only HET offspring (as these are the only genotype common to all three maternal genotypes) showed a similar effect, indicating a clear role for maternal genotype in the development of the pups. While the WT and HOM offspring genotypes were not further statistically analysed, a similar trend was seen with the WT pups from WT mothers being heavier than the WT from HET mothers and the HOM pups from HET mothers being heavier than the HOM pups from HOM mothers (see fig 3).

The changes in bodyweight can be explained by two different theories: The first possibility is a difference in prenatal development, while the second cause could be related to differences in postnatal maternal behaviour because of the genotype, or a combination as both theories are not mutually exclusive.

The most logical conclusion is that the postnatal environment causes this change. There is no indication of an offspring genotype effect in this experiment; this makes it less likely that SERT expression directly affects the bodyweight. If SERT expression would influence the bodyweight directly, a change in bodyweight based on offspring genotype would also be expected. Also, maternal serotonin is the main source of serotonin during a small period in the development of the body. Therefore, it is more likely that a different variable caused the results produced in this study.

Research in this lab has shown a significant reduction in pup retrieval in HOM mothers compared to HET and WT mothers (not published). Previous observations in the lab also indicate an increase in mortality in pups from HOM mothers caused by neglecting their pups (not published). This makes it more likely that the differences found are a result of reduced maternal care. However, given the setup of the experiment, conclusions cannot be conclusively drawn. To provide a reasonable

indication of the influence of the prenatal and the postnatal environments, the experiment should be repeated, but bodyweight should be measured over a longer time span, and weighing should start at PN 0. Alternatively, rats could be cross-fostered immediately after birth to selectively manipulate the postnatal environment.

Research into the influence of the SERT genotype on bodyweight is scarce. Homberg et al. examined the bodyweight at PN 21 and only found a significant difference in female rats (72). Previously unpublished research in this group found a genotype effect, a maternal effect and an interaction effect on bodyweight. It is necessary to take into consideration the fact that results of the previously unpublished study were very unbalanced, making interpretation thereof difficult.

The lack of difference between the groups could in part be an effect of gender. In this study, only male rats were analysed, whereas, in the study conducted by Homberg et al. only female rats showed a difference in weight at PN 21(72). Therefore the weight of female rats should be included in order to provide a full overview of the influence of genotype and maternal genotype on weight.

The analysis of litter size compared to maternal genotype and bodyweight did not show a significant difference. This analysis was not one of the main goals for the study; rather, it was executed as a possible hypothesis generating experiment. Thus, when interpreting the results of this analysis, it should be taken into consideration that the sample sizes were too small (3 litters per genotype) to identify relevant differences. It is not possible to reject the hypothesis that the SERT genotype influences litter size or that litter size influences the body weight at PN 7, so, to study this hypothesis further, a different experiment should be designed. However, the analysis does not provide a direct indication of the relevance of this aspect of the study.

Conclusion: Weight analysis

Based on the results of this experiment the conclusion can be drawn that maternal genotype influences the body development of rat pups. A reduction in SERT activity led to a reduction of the bodyweight of the rat pups at PN 7. Future studies will be required to determine if this effect is postnatal or prenatal and to identify its exact cause.

Discussion: Neurogenesis study

At the time of writing, the results cannot be discussed. The mistakes in the method are already described in the results paragraph.

Disadvantages of the methods used

One of the limitations of this study is the quality of the animals used in the experiment. The Animals used for breeding were previously used for other experiments, which could have caused changes in behaviour epigenetics that may have influenced the development of their offspring(73). There is no realistic way to prevent this from happening, as limitations in terms of space, time and money make the purchase of a naïve set of animals for every experiment unrealistic. However, in an ideal world, the animals, and their ancestors, would have been naïve to experiments. For this experiment, it was impossible to take into account the histories of the parents or grandparents. Therefore, possible epigenetic or behavioural effects from previous experiments could affect the behaviour towards the pups or the transfer of epigenetic changes. The number of mothers in this experiment was quite small; thus, one mother with behavioural issues as a result of previous experiments could have severely affected the outcome of this study(73,74).

One of the issues with the use of BrdU in animals is the injection procedure, as injecting the animals could possibly cause a stress response. The young pups were separated from their mothers, which alone could be a stressor, and experienced a very stressful event(67). Even though the pups were in the stress hypo-responsive period at the age of PN 7, it may still be possible that the exposure to injection caused some kind of response, which may have influenced neurogenesis. Especially because SERT^{-/-} rats are known to respond more intensely to stressful stimuli, the difference in neurogenesis measured could have been a consequence of a difference in stress (64). Future research could therefore benefit from using another marker for neurogenesis. For example, the ki-67 protein, this protein is naturally available in the brain and therefore could be measured without the interference of an injection. Another advantage of ki-67 over BrdU is that one variable in the process is eliminated. Ip injections in seven-day old rats are prone to complications due to of the size of the animals. Different injection sites could cause different levels of absorption and distribution in the body and therefore introduce an unnecessary uncertainty into the model(75).

The influence of the maternal effect is not as big as it possibly might be, as it was only possible to analyse its impact during a short interval in which only maternal behaviour could influence the neurogenesis. In this experiment an interval of two hours on PN 7 was chosen, BrdU doesn't mark the cells created during the prenatal period(67); therefore miswiring or an increase in cell density leading to behavioural deficits during prenatal development may not have been identified by the method used. To correctly evaluate the maternal effect the foetus should be analysed.

This study is the first to investigate the effect of a genetic increase in maternal serotonin on neurogenesis. However, it does not provide a full account of the exact role of maternal serotonin. Because this study was conducted using rats at PN 7, its findings could have been the result of differences not only serotonin concentrations but also in maternal behaviour. Previous studies in this group and research with tryptophan hydroxylase-2 (TPH2) knockout mice suggest a difference between KO and WT mothers in the maternal behaviour directed towards pups (76). KO mothers demonstrate less interest in their pups up to the point of neglecting them completely. This could cause a stress response in the pups and therefore cause changes, leading to different neurogenesis rates(77). A solution to at this phenomenon would be to repeat the experiment but using pups at PN 0. It's easier to stain the samples with a marker other than BrdU, because injections straight after birth could be difficult to execute. Therefore, marking with ki-67 to investigate the neurogenesis could be a more viable method.

One other problem with maternal behaviour is the problem that a mother may display different behaviours towards certain pups based on their genotype. No behavioural experiments were performed in this study, but it would seem logical that that when a mother behaves differently towards, for example, HOM pups when compared to WT pups, the HOM pups would experience this as being stressful and their brain development would therefore be altered. To investigate this principle a behavioural study should be conducted.

Implications for future research

Gender effects

A possible future research direction could be an extension of this study. In this research, only the males were taken into account; to investigate possible differences between males and females, a study intended to determine if there is a gender effect could be conducted. This would also provide greater insight into the discovered weight differences, as Homberg et al. found a weight difference on PN day 21 only in female SERT knockout rats when compared to WT rats; in males, this effect was not visible. The authors did not present any data concerning weight changes that may have occurred on PN day 7 (72). Also research with fluoxetine has shown differences between males and females in development related to changing serotonin concentrations. Therefore, it would be viable to repeat the experiment with females included(78).

Stressful events

Another possibility for investigating the influence of a combination of SERT genotype and an early life adverse event is the so-called gene x environment (GxE) study. There is no consensus regarding the existence of the GxE effect, but several studies have demonstrated an increase in depression and generalized anxiety disorder rates on the part of individuals who possess the 5-HTTLPR mutation and have experienced an early life adverse event (79). Stress has also been shown to alter cortical morphology in rats' early lives. To investigate this, a similar study could be designed, with an extra arm: Half of the rats studied could undergo an early life stress event or a prenatal stressor in order to create a GxE environment.

The early life stress event could be maternal separation for three hours a day from PN 2 up until PN 7. On PN 7 the brains could be taken and the same staining procedures could be executed. Beyond early life stress, prenatal stress has also been shown to influence serotonin concentrations and 5-HT turnover in the dam and to increase foetal brain levels of tryptophan and serotonin in rats (80).

Therefore, another arm, in which the dams undergo a prenatal stress paradigm, could be added to this study. In addition, combining the experiments from the prenatal and postnatal stress would provide an opportunity to examine the differences in the effects of the two stressors in combination with the SERT transporter.

Another possibility could be cross-fostering at PN 0. Cross-fostering has shown to reverse the behavioural effects of the maternal genotype in rats with changed brain serotonin concentrations due to a TPH2 mutation(76). Data concerning SERT^{-/-} animals are unavailable it could therefore be a useful experiment to fully assess the role played by the postnatal environment in neurogenesis.

Discovering the sort of cell

This study only considered the role of SERT in neurogenesis, but the role played by serotonin in the formation of the brain is far more complex. Previous studies have indicated that serotonin plays a role in cortical mapping and neuronal differentiation (8). However, how exactly this functions and the influence of the maternal genotype remain undiscovered. Analysis with BrdU alone does not contribute to unravelling this mystery either, as BrdU is a thymidine analogue and therefore only provides information concerning the formation of new DNA. It is a marker used to view the mitosis process, but it does not provide any indication of the phenotype of the formed cell. Furthermore, it marks newly formed neurons as well as glia cells (81). Therefore, even if there is no difference in neurogenesis rates, serotonin could also alter differentiation into different cell types and thus cause changes in brain development. A method for investigating phenotype is double marking with a second neuronal marker, such as doublecortin. Doublecortin is a marker used to investigate the differentiation of neuronal cells to neurons (82). An experiment using doublecortin should be executed differently to one that uses BrdU alone: Because differentiation takes a longer time after neurogenesis, the rat brains should be taken at a longer time interval after the injection of the BrdU. The ratio between doublecortin-stained cells and BrdU-stained cells provides an indication of the influence of serotonin on neuronal differentiation.

Interneuron density

Interneurons are connecting neurons in the human brain. They are the connections between sensory and motor neurons and are involved in the connections between different brain areas. The functioning of the brain is a very complicated and delicate process in which all brain areas are in perfect synchrony. One of the major influential factors that keeps the brain functioning is the connectivity between its different areas (7). Interneurons are crucial in this process; thus, reduced or increased interneuron density could cause behavioural deficits. Research shows that serotonin plays an important role in the migration of interneurons and connectivity. Reduced connectivity between the amygdala and the prefrontal cortex has been observed in ss- and sl-carriers (30). These results could be the product of a reduction in interneuron density. A staining technique that indicates interneuron density is parvalbumin staining. Parvalbumin is a protein presented by fast-spiking gabaergic neurons. A reduction in parvalbumin expression is an indication of a reduction of the number of interneurons (83). Genetic SERT overexpression in mice has been shown to impair the recruitment of parvalbumin neurons in the amygdala(84). The effect of reduced SERT expression is unknown. Further research into the parvalbumin neurons could provide insight into the role played by SERT mutation and maternal genotype in connectivity between brain cells.

Apoptosis

A future research direction could also be to investigate the effect of apoptosis on the rat brain. In this project, only the effect of neurogenesis was assessed, but neurogenesis is only a small part of the processes mediated by serotonin. Serotonin has been shown to affect concentrations of BDNF in the brain. The concentrations thereof are lowered in SERT^{-/-} rats when compared to SERT^{+/+} rats (46). BDNF is known to be a crucial factor in cell survival and apoptosis (7). A lack of BDNF could therefore influence the cortical morphology or intercellular connections of the SERT Knockout rats. A lack of BDNF could therefore influence the cortical morphology or intercellular connections of the SERT knockout rats. Terminal deoxynucleotidyl transferase dUTP nick end labelling (TUNEL)

presents a means of investigating this principle; this method examines DNA fragmentation and therefore provides an indication of cell apoptosis(85).

Spine density

Another function of serotonin in the developing brain is developing the appropriate morphology and connectivity of neuronal cells(8). Even if the number of cells remains the same between the SERT^{-/-} animals and the SERT^{+/+} animals, incorrect connectivity between cells could still be a cause of the behavioural differences observed between the genotypes. Increased spine density results in an increase in signal transmission and could therefore cause overactivity in certain brain areas. Nietzer et al. investigated spine density in SERT^{-/-} mice and found an increase in spine density compared to non-stressed animals (42). Golgi-Cox analysis provides a means of examining spine density and branching; this is an analytical staining method in which 3 to 5 percent of the total number of cells are randomly selected. The reason why Golgi Cox only selects 3 to 5 percent of the cells is unknown, but the result of the Golgi Cox staining provides a visualisation of neurons in their entirety, including cell soma, axons, dendrites and spines (86).

The influence of 5HT1a and BDNF

As previously described, serotonin causes a wide array of effects. Several studies showing the influence of the SERT on 5ht1a density and BDNF concentrations have been performed (46–48). These studies have shown a decrease in 5ht1a expression in the amygdala but not the hippocampus(87). Therefore, if the results would have shown an increase in neurogenesis in the hippocampus but not in the amygdala, the role of the 5ht1a receptor could be crucial. Furthermore, in the case of a clear maternal effect, the role of the 5ht1a receptor could be influential. For example, if a mother possesses a different SERT genotype to her pups, this could lead to desensitization of the 5ht1a receptor in early development. Therefore, neurogenesis could be changed at PN 7, resulting in different brain development. The marking of 5ht1a receptors could provide additional information regarding the maternal influence on neurogenesis. A means of doing this is with the RNA of the 5ht1a

receptor, as it can be measured in different brain areas and can be used to quantify 5ht1a expression. Brain-derived neurotrophic factor can be investigated in a similar fashion, as BDNF DNA can be quantified. Primers therefore were previously used by Calabrese et al. (46).

Conclusion

It is difficult to provide an answer to the original question based on the data presented previously; the missing data make drawing conclusions even more challenging.

A conclusion can be drawn about the influence of the SERT genotype. The offspring genotype does not affect the weight at PN 7. Nonetheless, the maternal genotype does, a reduction in SERT expression leads to a decrease in offspring bodyweight. The cause of this effect remains elusive, but future studies are presented in this thesis to discover the cause of the difference in bodyweight.

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